Program: **Diploma in Automation and Robotics**	
Course Code: **4332**	Course Title: **Microcontroller based Robotic Application Design**
Semester: **4**	Credits: **3**
Course Category: **Program Core**	
Periods per week: **3 (L:3, T:0, P:0)**	Periods per semester: **42**

Course Objectives:

- To teach students design and interfacing of microcontroller-based embedded systems.
- To understand concepts of 8051 Processor and programming them.
- Understand the key concepts of embedded systems such as I/O, timers, interrupts and interaction with peripheral devices.
- Understand Low-Level and Embedded C Programming.
- To analyze and implement various interfacing circuits necessary for various applications

Course Prerequisites:

Topic	Course Code	Course name	Semester
Number System, ADC/DAC		Analog and Digital Electronics	3
Sensors, Motors		Foundations of Robotics	2

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Describe the architecture of 8051 microcontrollers	10	Understanding
CO2	Familiarize and practice assembly language programming of 8051	10	Applying
CO3	To familiarize with embedded C language and practice Timer and Serial port programming using embedded C language.	10	Applying
CO4	To design some robotic applications by utilizing various 8051 modules and C language	10	Applying
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3	2					
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Describe the architecture of 8051 microcontrollers**		
M1.01	Understand basic concept of embedded systems structure	1	Understanding
M1.02	Comparative study of commonly used microcontroller family	1	Understanding
M1.03	Familiarize with the internal architecture of 8051 Microcontroller	4	Applying
M1.04	Familiarize with the memory organization of 8051	2	Applying
M1.05	Understand usage of various pins of 8051	2	Applying
Contents: Functional units of a Embedded System– Processors -CISC versus RISC Processors – Features of 8051, PIC, AVR and ARM Microcontrollers **-8051 Microcontroller architecture**: Introduction to MCS -51Family Micro-controllers, Architectural block Diagram, General Purpose and Special Function Registers, I/O Port circuits, Memory organization, Internal program and data memory – External Memory, Pin diagram and Pin Functions			
CO2	**Familiarize and practice assembly language programming of 8051**		
M2.01	Familiarize with the instruction set of 8051	4	Applying
M2.02	Understanding the concepts stack operations related to 8051	2	Applying
M2.03	Practice interfacing LED, LCD Display using	3	Applying

	assembly language		
	Series Test – I	1	
Contents: 8051 Assembly language programming: Programming model of 8051, Addressing modes, programming of 8051 based on data transfer, arithmetic and logical group, branching instructions, bit manipulation instructions and I/O Port programming. Concept of stack, subroutine and related instructions – **Sample Programs assembly language** – LED interfacing with delay -16x2 LCD Module interfacing – Keyboard Interfacing			
CO3	**To familiarize with embedded C language and practice Timer and Serial port programming using embedded C language.**		
M3.01	Familiarize with basic concepts of embedded C language	3	Applying
M3.02	Familiarize with 8051 timer operation and practicing timer programming	4	Applying
M3.03	Familiarize with 8051 serial port operation and practicing serial programming	3	Applying
Contents: Embedded C – Arithmetic – logical – Data transfer – Decision Control – Looping - **8051 Timers**- Associated registers - Various modes of timer/counter operations - Time delay programs in C - **8051 Serial Port and Programming**: Basics of serial communication, RS232 standards, 8051 connection to RS232, Serial data input/output and associated registers, Various modes of serial data communication, serial data communication programs in C			
CO4	**To design some robotic applications by utilizing various 8051 modules and C language**		
M4.01	To familiarize with 8051 interrupts and interrupt handling.	3	Understanding
M4.02	To practice the interfacing of some commonly used elements related to robotic applications	7	Understanding
	Series Test – II	1	
Contents: 8051 Interrupts: Concept of Interrupt, interrupt versus polling, Types of interrupts in 8051, Reset, interrupt control and associated registers, interrupt vectors, Interrupt execution, RETI instruction, software generated interrupt, interrupt handler subroutine for timer/counter and serial data transmission/reception inC,**Robotic application design** - Relay, PWM, DC and Stepper Motors: Relays and Opt-isolators, Stepper motor interfacing, DC motor interfacing and PWM using 8051 - Proximity Sensor Interfacing – 8 bit ADC/DAC Interfacing			

Text / Reference

T/R	Book Title / Author
T1	Muhammed Ali Mazidi& Janice GilliMazidi, R.D. Kinley, The 8051 microcontroller and Embedded System, Pearson Education, 2nd edition.
R1	The 8051 Microcontroller & Embedded Systems using Assembly and C by K. J. Ayala, D. V.Gadre (Cengage Learning, India Edition).
R2	The 8051 Microcontrollers: Architecture, Programming and Applications by K Uma Rao, AndhePallavi, Pearson Education.
R3	Michael Bass, "Programming Embedded Systems in C and C++", Oreilly, 2003
R4	8051 Microcontroller: Internals, Instructions, Programming and Interfacing by Subrata Ghoshal, Pearson Education.
R5	Michael J. Pont, "Embedded C", Addison Wesley, 2002.

Online Resources

Sl No	Website Link
1	https://www.tutorialspoint.com/microprocessor/microcontrollers_8051_architecture.htm
2	https://www.javatpoint.com/embedded-system-8051-microcontroller
3	https://www.geeksforgeeks.org/introduction-to-8051-microcontroller/
4	https://www.electronicsforu.com/technology-trends/learn-electronics/8051-microcontroller-basics
5	https://nptel.ac.in/courses/117/104/117104072/